

NEURALPLANE: STRUCTURED 3D RECONSTRUCTION IN PLANAR PRIMITIVES WITH NEURAL FIELDS

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We aim for structured plane-based 3D reconstruction

- We target clean, semantic planar primitives from casual multi-view capture.
- We find cross-view inconsistencies and shape–radiance ambiguity are the core challenges.
- We observe RANSAC pipelines miss recall; neural surfaces lack structure and semantics.

We fuse 2D plane cues into a unified 3D neural plane field

- We distill noisy 2D plane observations into a 3D neural representation.
- We jointly learn geometry and semantics without plane annotations.
- We extract crisp, semantically aligned planar primitives.

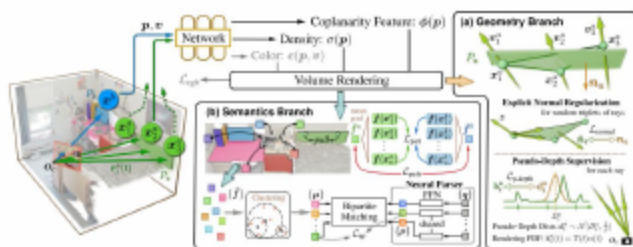


Figure: Plane-guided neural learning overview: geometry and semantics are trained jointly.

We generate 2D plane observations and initialize Local Planar Primitives

- We cluster monocular normals and intersect with SAM over-segmentation for smooth, semantic masks.
- We initialize plane normals by segment averages and offsets from weighted SfM keypoint distances.
- These LPPs provide strong constraints that guide learning.

We guide geometry with normal consistency and pseudo-depth

- We enforce explicit normal consistency from termination-point triplets to reduce shape–radiance ambiguity.
- We align rendered depth distributions to plane-derived pseudo-depth via KL and jointly refine plane parameters.

We learn coplanarity features and semantic prototypes

- We train a Neural Coplanarity Field with contrastive pull/push under geometric gating.
- We learn semantic prototypes online (Neural Parser) instead of brittle post-hoc clustering.
- We group termination points by learned prototypes and then fit global planes.



Figure: Neural Coplanarity Field visualization: view-consistent coplanar regions emerge.

We evaluate on ScanNetv2 and ScanNet++

- We show state-of-the-art geometry and segmentation versus strong baselines.
- We achieve balanced precision/recall and large gains in segmentation quality.
- We train per scene in minutes, enabling practical deployment.

We reconstruct crisp planes with coherent semantics

- We show clean planar structures and sharp edges across scenes.
- We maintain semantic consistency across views and through occlusions.

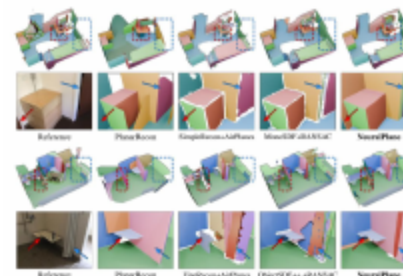


Figure: Qualitative results on ScanNetv2: crisp planes and aligned semantics.

- How can we better handle non-planar and mixed-planarity regions without over-planarizing?
- Can we reduce reliance on monocular/SfM priors and remain robust in the wild?
- How do we scale to large, cluttered, or outdoor scenes while preserving speed?
- Can we integrate edges/intersections and support online SLAM in real time?